Phase 1: Information and tasks

Status

Today is the 25th of October and all Scanlandian modelling teams have been assembled by their respective public health authorities to provide support for an ongoing outbreak of a new measles variant. It is suspected that the variant has originated in the US where there are informal reports of widespread outbreak activity, but US authorities have forbidden all data sharing with other countries.

A PCR-test has been developed for this new variant and as of today, 67 cases have been detected in Denden primarily clustered around the capital city of Vejager. One case have also been detected in Sweway.

The Scanlandian Health Authorities (SHA) suspect that the new mutated variant of measles escapes prior immunity to some (unknown) degree.

The symptoms appear similar to symptoms from the normal measles variant and some severe cases, requiring hospitalization, have been reported. Any more accurate assessment of severity has so far been difficult based on the fact that reported cases likely underestimate actual infections significantly.

Information and Data

The SHA have agreed to share line lists detailing all recorded cases from all four countries with the responsible modelling teams in the region. The line list is up-to-day as of the 25th of October.

The line list includes:

Variable	Type	Description
case_id	char	Person identifier
test_date	date	Date of test
symptom_onset_date	date	Date of symptom onset
country	char	Country of residence
region	char	Region of residence within country
age	integer	Age at symptom onset
vaccination_status	char	Vaccination status: unvaccinated, 2 doses, not offered (born to early)
hospital_admission_date	date	Date of hospital admission or blank
hospital_discharge_date	date	Date of hospital discharge or blank
date_of_death	date	Date of death or blank

Additionally, already existing data on demographics for each country and an estimate of daily travelers between the countries exists and can be used.

All data for this exercise can be found at: https://github.com/Gulfa/modelling_prep_ex/tree/main/exercise_data

Tasks

SHA have asked the modelling teams to provide input to a rapid risk assessment for their specific country taking into account all the information available.

- Is it likely that the measles outbreaks that we are observing will end in the near future?
- What can be said about the new variant of measles?
    - estimate how quickly the outbreak is growing
    - what can we say about disease severity?
    - What is the degree of immune escape?
    - How many new hospital admissions can we expect in each of the next four weeks?
        - How certain can we be about these estimates?
- Do you have any input for additional data that should be collected or shared?

    At 15:50 there will be a consensus plenum discussion of the above questions with an aim to reach a consensus about what we know and don't know at this time. Each group should arrive at the meeting ready to present their findings.

- Start thinking about making a planning scenario for the coming 6 months. Public health authorities will need to plan for how serious this outbreak can become. In a planning scenario, public health authorities want to understand the most likely trajectory and the reasonable best and worst case. The primary focus is on health care needs, especially hospitalizations, and a measure of the overall burden of infections and deaths.

Map of Scanlandia

